

The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for
Social Sciences

Tutorial Sheet 7- Trigonometric functions

July, 2023.

1. Find the quadrant that contains the terminal (end) side of θ if the given conditions are satisfied:

(a) $\sin \theta > 0$ and $\tan \theta < 0$	(d) $\tan \theta > 0$ and $\sec \theta < 0$
(b) $\sin \theta < 0$ and $\cos \theta < 0$	(e) $\sin \theta < 0$ and $\cot \theta < 0$
(c) $\cos \theta > 0$ and $\tan \theta < 0$	(f) $\csc \theta > 0$ and $\sec \theta > 0$
2. Without the use of a calculator, convert each of the following to radians:

(a) -250°	(c) 75°	(e) 570°
(b) 25°	(d) -330°	(f) 635°
3. Without the use of a calculator, convert each of the following to degrees:

(a) $\frac{7\pi}{3}$	(c) $\frac{\pi}{13}$	(e) $\frac{7\pi}{6}$
(b) $-\frac{3\pi}{5}$	(d) $\frac{5\pi}{3}$	(f) $-\frac{11\pi}{4}$
4. Without using a calculator or table, find the exact values of each of the following:

(a) $\sin 75^\circ$	(d) $\sin(-135)^\circ$	(g) $\cot(-\frac{5\pi}{3})$
(b) $\cos(-15)^\circ$	(e) $\tan 315^\circ$	(h) $\sec \frac{7\pi}{6}$
(c) $\cos 150^\circ$	(f) $\cos \frac{7\pi}{3}$	(i) $\csc(\frac{\pi}{12})$
5. Given that θ is an angle in the third quadrant and $\cos \theta = \frac{-4}{5}$. Find $\cot \theta$ and $\csc \theta$.
6. Simplify each of the following to a single trigonometric function or a constant:

$$\begin{array}{lll}
 \text{(a)} \sec x - \sin x \tan x & \text{(c)} \cos x + \tan x \sin x & \text{(e)} \frac{\sec \theta - \cos \theta}{\tan \theta} \\
 \text{(b)} \csc x - \cos x \cot x & \text{(d)} (\sin^2 x - 1)(\tan^2 x + 1) & \text{(f)} \frac{\sec \theta + 1}{\tan \theta + \sin \theta}
 \end{array}$$

7. Prove each of the following identities:

$$\begin{array}{ll}
 \text{(a)} \tan^2 x + 1 + \tan x \sec x = \frac{1 + \sin x}{\cos^2 x} & \text{(e)} \frac{(\tan x)(1 + \cot^2 x)}{1 + \tan^2 x} = \cot x \\
 \text{(b)} \frac{\cos x + \sin x}{\cos x} = 1 + \tan x & \text{(f)} \frac{1 - \sin x}{1 + \sin x} = (\tan x - \sec x)^2 \\
 \text{(c)} \frac{\csc x - 1}{\cot x} = \frac{\cot x}{\csc x + 1} & \text{(g)} \frac{1}{\sec x(1 - \sin x)} = \sec x + \tan x \\
 \text{(d)} \frac{1}{\tan x + \cot x} = \cos x \sin x & \text{(h)} \frac{\tan x + \tan y}{\cot x + \cot y} = \tan y \tan x
 \end{array}$$

8. Verify each of the following:

$$\begin{array}{lll}
 \text{(a)} \cos(2x) = 2\cos^2 x - 1 & \text{(c)} \cos(\theta - \pi) = -\cos \theta & \text{(e)} \sin(\theta + \frac{\pi}{2}) = \cos \theta \\
 \text{(b)} \cos(2x) = 1 - 2\sin^2 x & \text{(d)} \cos(\theta + \frac{\pi}{2}) = -\sin \theta & \text{(f)} \tan(\theta - \pi) = \tan \theta
 \end{array}$$

9. Without using a calculator or a table, solve each of the following equations for $0 \leq x \leq 2\pi$:

$$\begin{array}{lll}
 \text{(a)} 2\sin x + 1 = 0 & \text{(c)} 2\cos^2 x - \sqrt{3}\cos x = 0 & \text{(e)} 2\cos^2 x + \cos x = \sin^2 x \\
 \text{(b)} \sec x = 2 & \text{(d)} \sin x + 2 = 3 & \text{(f)} \tan x = \cot x
 \end{array}$$

10. Without using a calculator or a table, solve each of the following equations for $0 \leq x \leq 4\pi$:

$$\begin{array}{ll}
 \text{(a)} 2\tan x \sec x - \tan x = 0 & \text{(c)} \cos 2x + 3\sin x - 2 = 0 \\
 \text{(b)} \sec^2 x - \sec x - 2 = 0 & \text{(d)} \sin x + \cos x = 1
 \end{array}$$

11. Sketch each of the following in the indicated interval:

$$\begin{array}{ll}
 \text{(a)} f(x) = 1 + \sin x, \quad 0 \leq x \leq 2\pi & \text{(c)} f(x) = -\cos x, \quad 0 \leq x \leq \frac{2\pi}{3} \\
 \text{(b)} f(x) = 1 - \cos x, \quad 0 \leq x \leq 2\pi & \text{(d)} f(x) = \frac{1}{2} \sin\left(\frac{1}{2}x\right), \quad 0 \leq x \leq 4\pi
 \end{array}$$

12. Find the period, amplitude and phase shift of each of the following functions, and in each case sketch 1 complete revolution for $x \geq 0$.

$$\begin{array}{lll}
 \text{(a)} f(x) = 2\sin 2x & \text{(d)} f(x) = \cos(x + 15^\circ) & \text{(g)} f(x) = 2 + 3\cos(2x - \frac{\pi}{2}) \\
 \text{(b)} f(x) = \cos(2x - \frac{\pi}{2}) & \text{(e)} f(x) = -4\cos(3x) & \\
 \text{(c)} f(x) = 3\sin(x + 30^\circ) & \text{(f)} f(x) = \frac{2}{3}\sin 2(x - \pi) &
 \end{array}$$